

Lebanese University

Faculty of Engineering III

Beyrouth-hadath



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Concurrent Programming Course

Presented for: Dr. Mohammad Aoude

Prepared by : Assil Shehade 6704

Ali Ataya 6756

Mohammad Hammoud 6460

Mohammad Salloum 6494

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➤ Chapter 1 — Introduction

1.1 Project Overview

The CedrusTech HR System is a distributed Human Resources management platform designed to support both candidate recruitment and employee interaction through artificial intelligence technologies. The system combines modern web technologies, concurrent programming techniques, and large language model (LLM) capabilities to provide real-time communication and automated recruitment services.

The project was developed using a multi-tier architecture consisting of a Java Spring Boot backend, a Python FastAPI AI service, a SQL Server database, and a browser-based frontend implemented with HTML, CSS, and JavaScript. The system integrates an AI-powered chatbot capable of answering company-related questions using Retrieval-Augmented Generation (RAG) techniques and a job application portal that allows candidates to submit applications online.

A key objective of the project is to demonstrate the practical application of concurrent programming concepts in a real-world distributed system. Unlike traditional web applications that process requests sequentially, the CedrusTech HR System is designed to handle multiple users simultaneously while maintaining responsiveness, reliability, and fault tolerance.

The Java backend serves as the concurrency engine of the system. It manages WebSocket sessions, thread pools, synchronization mechanisms, asynchronous task execution, backpressure controls, metrics collection, distributed tracing, and graceful shutdown procedures. The Python backend focuses exclusively on artificial intelligence operations, including vector retrieval using ChromaDB and response generation using the Ollama Llama 3 large language model.

The final architecture separates business logic, concurrency management, and AI processing into independent components, allowing the system to scale while maintaining clear responsibilities between services.

1.2 Problem Statement

Modern HR departments increasingly require digital platforms capable of handling large numbers of applicants while providing immediate responses to candidate inquiries. Traditional HR systems typically rely on synchronous request processing, where each

request occupies server resources until completion. As user traffic increases, these systems often experience bottlenecks, increased latency, and reduced availability.

Several challenges were identified during the design phase of the CedrusTech HR System:

Challenge 1: Simultaneous User Access

The chatbot service must support multiple users communicating with the system concurrently. A slow AI response for one user should not delay responses for other connected users.

Challenge 2: Long-Running AI Operations

Generating responses using large language models can take several seconds. Blocking server threads while waiting for AI responses would significantly reduce overall system throughput.

Challenge 3: Resource Protection

Uncontrolled request growth can exhaust memory and thread resources. The system requires mechanisms that prevent individual users from monopolizing shared resources.

Challenge 4: Fault Tolerance

External AI services may become unavailable, experience network failures, or exceed response time limits. The system must recover gracefully without affecting other users.

Challenge 5: Real-Time Communication

The chatbot requires bidirectional communication between the browser and the server. Traditional HTTP polling introduces unnecessary overhead and latency.

Challenge 6: Application Persistence

Job applications and uploaded CV files must be stored reliably while ensuring duplicate submissions are detected and rejected.

To address these challenges, the system adopts a concurrency-focused architecture built around asynchronous processing, bounded thread pools, synchronization primitives, event logging, distributed tracing, and failure recovery mechanisms.

1.3 Project Objectives

The primary objective of the CedrusTech HR System is to develop a reliable and scalable HR platform that demonstrates advanced concurrent programming concepts through practical implementation.

Specific objectives include:

Real-Time AI Chatbot

Develop a chatbot capable of providing instant responses to user questions through WebSocket-based communication and Retrieval-Augmented Generation techniques.

Concurrent Request Processing

Allow multiple clients to interact with the system simultaneously without blocking one another or causing performance degradation.

Scalable Resource Management

Implement bounded thread pools, queue management strategies, and backpressure mechanisms that prevent resource exhaustion under heavy load.

Fault-Tolerant Operation

Ensure the system continues operating correctly when external services become unavailable or experience transient failures.

Distributed Tracing

Provide end-to-end visibility of requests across Java and Python services using correlation identifiers and event logging.

Graceful Shutdown

Allow the application to terminate safely without losing active requests, database writes, or WebSocket sessions.

Performance Monitoring

Collect operational metrics that allow developers and administrators to monitor throughput, latency, queue utilization, and overall system health.

1.4 Scope of the Project

The scope of the CedrusTech HR System includes both recruitment management and AI-assisted communication services.

The implemented system provides:

- Real-time AI chatbot functionality
- Job application submission

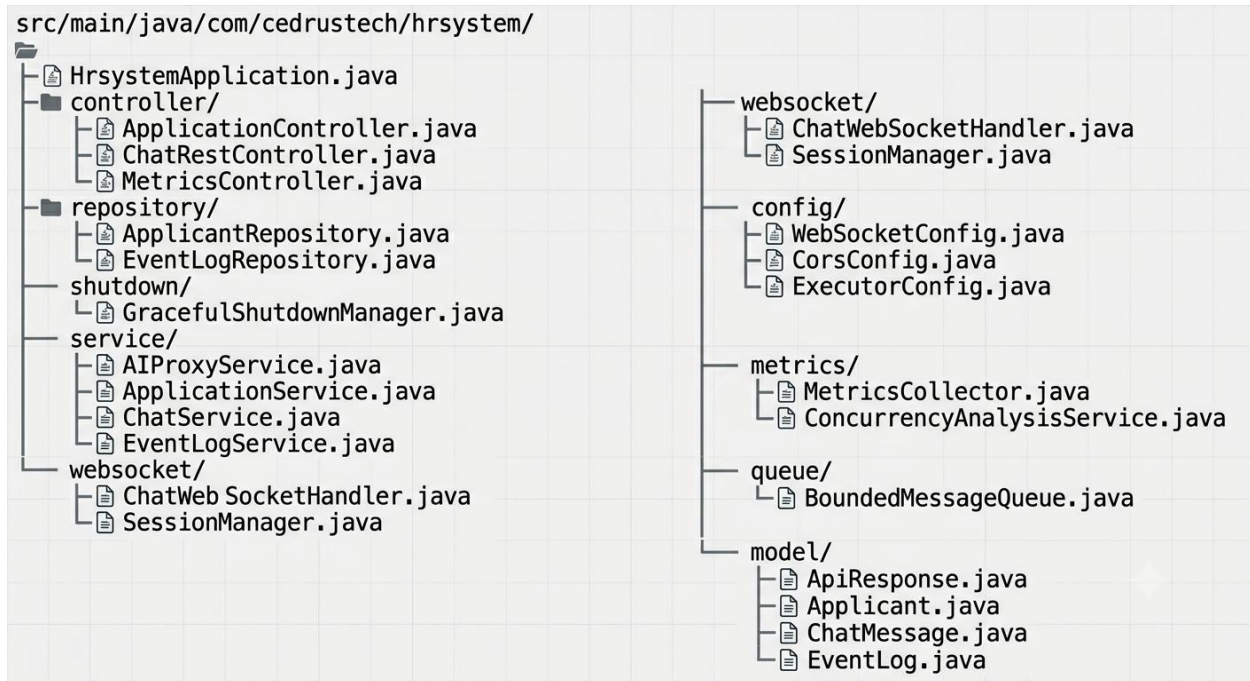
- CV file storage
- Applicant persistence in SQL Server
- Event logging and auditing
- Distributed request tracing
- Performance monitoring
- Graceful shutdown support
- Concurrent client handling

Syllabus Compliance Summary

Week(s)	Requirement	Implementation
1–2	Concurrent entry point, thread lifecycle	WebSocket /ws/chat + HTTP /apply + /chat
3	Race conditions, monitors, locks, deadlock avoidance	ConcurrentHashMap, synchronized(session), ReadWriteLock
4	Visibility, happens-before, volatile, immutability	volatile maintenanceMode, ChatMessage/ApiResponse records
5	ExecutorService, bounded queues, backpressure, atomics, RW locks	3 ThreadPoolExecutors, LinkedBlockingQueue, BackpressureRejectionHandler, LongAdder
6	Fork/Join, task granularity, break-even measurement	ForkJoinPool CV analysis, 500-word threshold with timing data
7	CompletableFuture, async pipelines, timeouts, fallbacks, graceful shutdown	CompletableFuture chain, orTimeout(30s), GracefulShutdownManager
9	Python GIL, runtime limits	aiExecutor max=5, Semaphore(5), circuit breaker, HTTP/1.1 forced
10–14	Network failure, event logs, idempotency, tracing, failure recovery	event_log table, correlationId chain, isAlreadyDelivered(), retry with backoff

All requirements are satisfied. No unbounded thread creation or unbounded queues exist anywhere in the system. Overload behavior is intentional (backpressure, circuit breaker) and measurable (metrics endpoint). The event log enables both idempotency and distributed tracing across the Java–Python boundary.

1.5 Project Directory Structure: com.cedrustech.hrssystem



1.6 Report Organization

This report is organized according to the concurrent programming concepts covered throughout the course.

Chapter 2 presents the overall system architecture and explains how requests flow through the application.

Subsequent chapters examine synchronization mechanisms, thread management, executor services, asynchronous processing, fault tolerance, distributed tracing, metrics collection, and graceful shutdown procedures.

Each chapter maps the implemented solution to specific concurrent programming principles and demonstrates how theoretical concepts were applied in a practical software engineering project.

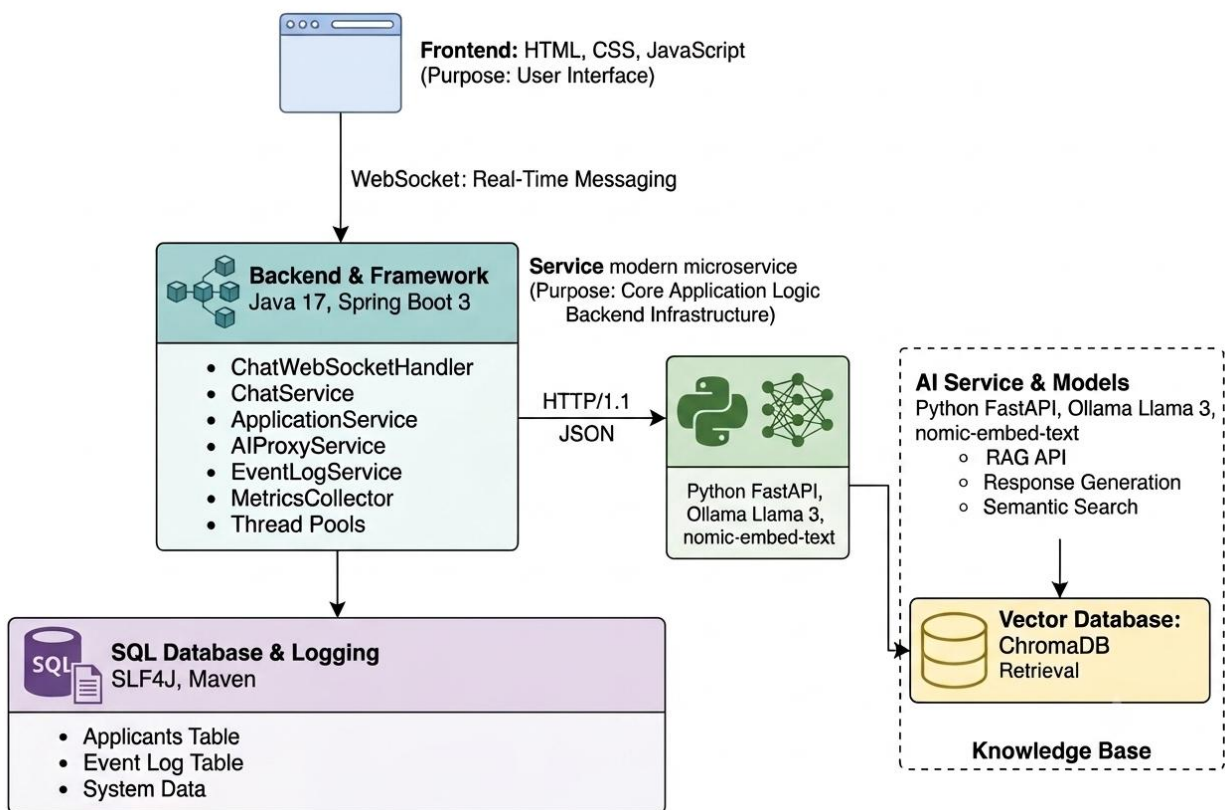
➤ Chapter 2 — System Architecture

2.1 Architectural Overview

The CedrusTech HR System follows a layered distributed architecture consisting of four major components:

1. Browser Frontend
2. Java Spring Boot Backend
3. Python FastAPI AI Service
4. SQL Server Database

The architecture separates user interaction, concurrency management, artificial intelligence processing, and persistent storage into independent layers.



This separation ensures that concurrency management remains centralized within the Java backend while computationally intensive AI tasks are isolated inside the Python service.

2.2 Technology Stack

The system combines several technologies, each selected for a specific responsibility within the architecture.

Layer		Technology	Purpose
Frontend		HTML, CSS, JavaScript	User Interface
Communication		WebSocket	Real-Time Messaging
Backend		Java 17	Core Application Logic
Framework		Spring Boot 3	Backend Infrastructure
AI Service		Python FastAPI	RAG API
Language Model		Ollama Llama 3	Response Generation
Vector Database		ChromaDB	Document Retrieval
Embedding Model		nomic-embed-text	Semantic Search
Database		SQL Server	Persistent Storage
Logging		SLF4J	Application Logging
Build Tool		Maven	Dependency Management

The chosen technologies provide a balance between performance, maintainability, and scalability while supporting the concurrency requirements of the project.

2.3 Frontend Architecture

The frontend is implemented using standard web technologies and communicates exclusively with the Java backend.

The browser establishes a persistent WebSocket connection with the server on port 8081. Unlike traditional request-response communication, WebSockets provide full-duplex communication, allowing the server to send AI responses immediately after they become available.

The JavaScript client performs three major functions:

1. Establishes WebSocket connections.

2. Sends chat messages.
3. Submits job application forms.

All chatbot traffic is routed through the Java backend. The browser never communicates directly with the Python AI service. This design centralizes concurrency management and simplifies security and monitoring.

2.4 Backend Architecture

The Java backend serves as the central processing layer of the system.

Several services collaborate to process incoming requests:

ChatWebSocketHandler

Receives WebSocket messages from browser clients and submits asynchronous tasks to executor services.

ChatService

Performs business logic processing, maintenance checks, greeting shortcuts, and AI request routing.

AIProxyService

Acts as a bridge between Java and Python, applying concurrency limits, retries, circuit breaker protection, and HTTP communication.

ApplicationService

Processes job applications, stores CV files, validates duplicate submissions, and persists applicants to SQL Server.

EventLogService

Records significant events asynchronously to support auditing, idempotency, and distributed tracing.

MetricsCollector

Maintains system performance metrics using lock-free data structures.

Together these components form the concurrency layer responsible for coordinating all system activities.

2.5 Request Processing Flow

Every chatbot message follows a structured processing pipeline.

Step 1 — Browser Submission

A user enters a message and sends it through the WebSocket connection.

Step 2 — WebSocket Reception

ChatWebSocketHandler receives the message and validates the session.

Step 3 — Queue Admission

The message enters a BoundedMessageQueue associated with the user session.

Step 4 — Executor Submission

A task is submitted to the wsExecutor thread pool through a CompletableFuture pipeline.

Step 5 — Business Logic Processing

ChatService performs maintenance checks and determines whether the request requires AI processing.

Step 6 — AI Request

AIProxyService sends an HTTP request to the FastAPI backend.

Step 7 — Retrieval-Augmented Generation

The Python service retrieves relevant documents from ChromaDB and generates a response using Ollama Llama 3.

Step 8 — Response Delivery

The generated answer is returned to Java and transmitted back to the browser through WebSocket.

Step 9 — Event Logging

EventLogService records processing events asynchronously in SQL Server.

This pipeline ensures that expensive AI operations never block network threads or other client sessions.

2.6 Job Application Workflow

The job application subsystem follows a separate processing path.

1. User completes the application form.
2. Browser sends HTTP POST request to /apply.
3. ApplicationService validates applicant information.
4. Duplicate email checks are performed.
5. CV file is stored on disk.
6. Applicant information is saved to SQL Server.
7. Event logs are written asynchronously.
8. Application status is set to **Pending**.

The submitted CV is later analyzed by a separate HR AI Agent operating outside the real-time request path.

➤ **Chapter 3 — Concurrent Entry Points and Thread Lifecycle**

3

.1 Introduction

O

ne of the primary goals of the CedrusTech HR System is the ability to serve multiple users simultaneously without performance degradation. Achieving this objective requires the system to support concurrent request processing while ensuring that long-running operations do not block other users.

Unlike a traditional sequential application, where each request must finish before another can begin, the CedrusTech HR System allows many independent operations to execute concurrently. This capability is particularly important because chatbot requests may require several seconds to complete due to AI processing performed by the Python backend and the Ollama language model.

To support concurrency, the system exposes multiple entry points that operate independently and are processed asynchronously through dedicated thread pools and executor services.

3.2 Concurrent Entry Points

The system exposes three major entry points that can receive requests concurrently from multiple clients.

Entry Point	Protocol	Purpose
/ws/chat	WebSocket	Real-time AI chatbot
/apply	HTTP POST	Job application submission
/chat	HTTP POST	REST chatbot endpoint

Each entry point serves a different business function while sharing the same concurrency infrastructure.

3.2.1 WebSocket Chat Endpoint

The primary communication mechanism of the chatbot is the WebSocket endpoint:

```
ws://localhost:8081/ws/chat
```

WebSockets provide a persistent full-duplex connection between the browser and the server. Unlike HTTP requests, which require a new connection for every interaction, WebSockets remain open throughout the conversation.

This design offers several advantages:

- Reduced communication overhead
- Lower latency
- Real-time response delivery
- Efficient server resource utilization
- Continuous bidirectional communication

Each browser client receives its own independent WebSocket session. Consequently, one slow user cannot block or interfere with other active users.

When a message arrives, the WebSocket handler immediately delegates processing to a background executor thread and returns control to Tomcat.

This approach prevents network threads from being blocked by expensive AI operations.

3.2.2 Job Application Endpoint

The job application endpoint allows candidates to submit resumes and personal information.

```
POST /apply
```

The endpoint accepts:

- First name
- Last name
- Email
- Phone number
- Position identifier
- Resume text

- CV file

Application processing involves several operations:

1. Validation
2. Duplicate email detection
3. File storage
4. Database persistence
5. Event logging

These operations involve both disk I/O and database I/O, which can be significantly slower than memory operations.

To avoid blocking request threads, application processing is delegated to the application's executor infrastructure.

The implementation is handled by the `ApplicationService` class.

The service generates a unique correlation identifier for every application:

```
<> Java  
  
String correlationId = UUID.randomUUID().toString();
```

This identifier is used throughout the system for tracing and auditing purposes.

3.2.3 REST Chat Endpoint

In addition to WebSocket communication, the system provides a traditional HTTP interface for chatbot access.

```
POST /chat
```

This endpoint is primarily used for testing, debugging, and external integrations.

Unlike the WebSocket implementation, the REST endpoint follows a request-response model.

Because AI responses can take several seconds to generate, the endpoint uses asynchronous processing techniques to avoid blocking Tomcat worker threads.

This allows the server to continue accepting new requests while AI processing occurs in the background.

3.3 WebSocket Session Lifecycle

Every WebSocket connection follows a well-defined lifecycle managed by the SessionManager component.

The lifecycle consists of four major stages:

1. Session Creation
2. Message Processing
3. Error Handling
4. Session Termination

3.3.1 Session Creation

When a client establishes a connection, Spring Boot invokes:

```
</> Java  
afterConnectionEstablished()
```

During this phase, several initialization tasks occur:

Session Registration

The session is stored in a concurrent session registry:

```
</> Java  
ConcurrentHashMap<String, WebSocketSession>
```

This structure enables lock-free access by multiple threads.

Queue Creation

A dedicated BoundedMessageQueue is created for the session.

Each client therefore receives its own isolated message queue.

Chat History Initialization

Conversation history is allocated and associated with the session.

Metrics Update

The active client counter is incremented.

At the end of this phase, the client is fully registered and capable of sending messages.

3.3.2 Message Processing

When a message arrives, Spring Boot invokes:

```
</> Java  
handleTextMessage()
```

The WebSocket handler performs several operations:

Message Validation

Incoming text is validated and normalized.

Correlation ID Generation

A unique request identifier is generated:

```
</> Java  
UUID.randomUUID()
```

This identifier follows the request throughout the system.

Event Logging

A CHAT_RECEIVED event is written asynchronously.

Queue Admission

The message is inserted into the session's bounded queue.

Executor Submission

Processing is delegated to the wsExecutor thread pool through a CompletableFuture.

The network thread immediately returns and is available to process new messages.

This design ensures that Tomcat threads never wait for AI responses.

3.3.3 Error Handling

Network interruptions and unexpected failures are inevitable in distributed systems.

Spring Boot invokes:

```
</> Java  
  
handleTransportError()
```

whenever a WebSocket communication failure occurs.

Examples include:

- Lost internet connection
- Browser termination
- Network timeout
- Protocol violations

The handler performs several recovery actions:

- Records failure metrics
- Logs the exception
- Cleans session resources
- Releases queue memory

This prevents resource leakage and ensures system stability.

3.3.4 Session Termination

When a connection closes normally, Spring Boot invokes:

```
</> Java  
  
afterConnectionClosed()
```

The session manager performs cleanup operations:

Session Removal

The session is removed from the registry.

Queue Cleanup

Associated message queues are released.

History Removal

Conversation history is discarded.

Metrics Update

The active client count is decremented.

The cleanup process prevents memory accumulation caused by inactive sessions.

3.4 Thread Lifecycle in CedrusTech HR System

Every task processed by the system follows a thread lifecycle consisting of four phases:

1. Thread Creation
2. Task Assignment
3. Task Execution
4. Thread Reuse or Termination

3.4.1 Thread Creation

Threads are created by Spring-managed ThreadPoolExecutor instances.

The system uses three dedicated executors:

Executor	Purpose
wsExecutor	WebSocket processing
aiExecutor	Python communication
appExecutor	Applications and logging

Threads are created only when needed.

Idle threads are reused to reduce creation overhead.

3.4.2 Task Assignment

When work arrives, it is encapsulated as a task and submitted to an executor.

Example:

```
</> Java

CompletableFuture.supplyAsync(
    () -> chatService.processMessage(message),
    wsExecutor
);
```

The executor selects an available worker thread.

If no thread is available, the task enters a bounded queue.

3.4.3 Task Execution

The worker thread executes the assigned task.

Examples include:

- Processing chat messages
- Calling the Python backend
- Saving applications
- Writing event logs

Execution occurs independently of the original network thread.

This separation is fundamental to the scalability of the system.

3.4.4 Thread Reuse

After completing a task, the thread is returned to the executor's pool.

Instead of being destroyed, it waits for another task.

Benefits include:

- Reduced CPU overhead
- Lower memory allocation costs
- Improved throughput
- Faster response times

Thread reuse is one of the main reasons thread pools outperform manual thread creation.

3.5 Avoiding Unnecessary Concurrency

An important principle of concurrent programming is that concurrency should only be introduced when it provides measurable benefit.

The CedrusTech HR System deliberately avoids creating asynchronous tasks for trivial operations.

For example, the ChatService contains a greeting shortcut:

```
<> Java

if (userText.matches("(?i)^(hi|hello|hey|hii|greetings)$")) {
    return "Hello! Welcome to CedrusTech Solutions.";
}
```

In this case:

- No executor task is created
- No HTTP request is sent to Python
- No AI processing occurs

The response is generated immediately within the Java layer.

This optimization reduces thread utilization and improves overall throughput.

3.6 Summary

This chapter examined the concurrent entry points and thread lifecycle of the CedrusTech HR System.

The system exposes three independent entry points: WebSocket chat, REST chat, and job application submission. Each request is processed asynchronously through dedicated executor services, ensuring that long-running operations never block network threads.

The WebSocket lifecycle includes session creation, message handling, error recovery, and resource cleanup, while thread pools manage task execution efficiently through thread reuse and bounded queues.

These mechanisms provide the foundation for the synchronization, locking, and visibility techniques discussed in the next chapter.

➤ Chapter 4 — Synchronization, Shared State, Visibility and Immutability

4.1 Introduction

Concurrent applications execute multiple threads simultaneously, often accessing the same data structures and shared resources. Without proper synchronization mechanisms, these threads may interfere with one another, producing inconsistent results known as race conditions.

The CedrusTech HR System processes multiple WebSocket messages, application submissions, database operations, and AI requests concurrently. Because many threads operate at the same time, shared state must be protected carefully to ensure correctness, consistency, and thread safety.

Different shared resources require different synchronization strategies. Instead of using a single locking mechanism everywhere, the system selects the most appropriate solution based on the access pattern and performance requirements of each component.

The primary synchronization mechanisms used in the project are:

- `ConcurrentHashMap`
- `ReadWriteLock`
- volatile variables
- synchronized blocks
- `AtomicInteger`
- `AtomicLong`
- `LongAdder`
- Immutable Java Records

Together these mechanisms prevent race conditions while maintaining high throughput under concurrent workloads.

4.2 Shared Mutable State in the System

Several components maintain state that can be accessed simultaneously by multiple threads.

Event Log Records Multiple background tasks Database transaction isolation

Shared Resource 	Accessed By	Synchronization Mechanism
Active WebSocket Sessions	Multiple handler threads	ConcurrentHashMap
Session Queues 	Multiple handler threads	ConcurrentHashMap
Maintenance Mode Flag 	ChatService readers/writers	ReadWriteLock + volatile
WebSocket Output Stream	Multiple executor threads	synchronized(session)
Metrics Counters 	All executors	LongAdder / AtomicLong
Circuit Breaker State 	AIProxyService threads	AtomicInteger
Event Log Records 	Multiple background tasks	Database transaction isolation

4.3 Race Conditions

A race condition occurs when multiple threads access shared data simultaneously and at least one thread modifies that data.

Consider a simple example:

```
</> Java
```

```
counter++;
```

Although this statement appears to be a single operation, it actually consists of three separate steps:

1. Read current value
2. Increment value
3. Write value back

If two threads execute these steps simultaneously, one update can overwrite the other.

For example:

```
Thread A reads 10  
Thread B reads 10
```

```
Thread A writes 11  
Thread B writes 11
```

Expected value:

12

Actual value:

11

One increment has been lost.

The CedrusTech HR System prevents similar issues by protecting all shared mutable state using thread-safe data structures and synchronization primitives.

4.4 ConcurrentHashMap

The most frequently accessed shared structure in the system is the session registry maintained by SessionManager.

Multiple threads may:

- Add new sessions
- Remove disconnected sessions
- Look up active sessions
- Access per-session queues

Using a traditional HashMap would be unsafe because concurrent modifications could corrupt the internal structure.

Instead, the system uses:

⌕ Java

```
ConcurrentHashMap<String, WebSocketSession>
```

And

⌕ Java

```
ConcurrentHashMap<String, BoundedMessageQueue>
```

ConcurrentHashMap provides thread-safe operations without requiring a global lock.

Benefits include:

- Concurrent reads
- Concurrent writes
- High scalability
- Minimal contention

Unlike Hashtable, ConcurrentHashMap locks only small portions of the structure when necessary.

This significantly improves performance when dozens of clients are connected simultaneously.

Why ConcurrentHashMap?

A synchronized HashMap would force every thread to wait for a single lock.

For example:

```
Client A wants to read Session 1
Client B wants to read Session 2
```

With a synchronized map:

```
Client B must wait
```

With ConcurrentHashMap:

```
Both operations proceed concurrently
```

This improves throughput under heavy load.

4.5 ReadWriteLock in ChatService

One important piece of shared state is the maintenance mode flag inside ChatService.

Administrators can enable maintenance mode while users continue sending requests.

This creates a classic reader-writer problem:

- Thousands of threads read the flag
- Very few threads modify the flag

Using synchronized would create unnecessary contention because every reader would block every other reader.

Instead, ChatService uses:

```
</> Java

private final ReadWriteLock lock =
    new ReentrantReadWriteLock();
```

4.5.1 Read Operations

Normal message processing acquires the read lock:

```
</> Java

lock.readLock().lock();
try {
    if (maintenanceMode) {
        return maintenanceMessage;
    }
} finally {
    lock.readLock().unlock();
}
```

Multiple readers can execute this code simultaneously.

Example:

```
Thread A → read
Thread B → read
Thread C → read
```

All proceed concurrently.

No blocking occurs.

4.5.2 Write Operations

Maintenance updates require exclusive access:

`</> Java`

```
lock.writeLock().lock();
try {
    maintenanceMode = enabled;
} finally {
    lock.writeLock().unlock();
}
```

Only one writer can execute at a time.

During a write:

```
Readers blocked
Writers blocked
```

This guarantees consistency.

4.5.3 Advantages Over synchronized

If synchronized had been used:

`</> Java`

```
synchronized(this) {
    if (maintenanceMode) ...
}
```

every read would acquire the same monitor.

Result:

```
Reader A blocks Reader B
Reader B blocks Reader C
```

Even though all operations are only reading.

ReadWriteLock eliminates this bottleneck and provides significantly better scalability.

4.6 Volatile Variables and Visibility

Synchronization is not only about atomicity.

Threads must also see the most recent values written by other threads.

Consider:

```
⟨⟩ Java
```

```
private boolean maintenanceMode;
```

Without synchronization, the JVM may cache this value inside a CPU register.

Example:

```
Admin Thread:  
maintenanceMode = true
```

Another thread may still see:

```
maintenanceMode = false
```

because it reads a stale cached copy.

To prevent this issue, ChatService declares:

```
⟨⟩ Java
```

```
private volatile boolean maintenanceMode;
```

And

```
⟨⟩ Java
```

```
private volatile String maintenanceMessage;
```

4.6.1 Happens-Before Relationship

The volatile keyword creates a happens-before relationship.

A write:

```
⟨⟩ Java
```

```
maintenanceMode = true;
```

becomes immediately visible to all threads.

Any subsequent read is guaranteed to observe the latest value.

This prevents stale data and visibility bugs.

4.6.2 Why Both volatile and ReadWriteLock?

Strictly speaking, ReadWriteLock already provides visibility guarantees.

However, using both mechanisms offers additional clarity:

ReadWriteLock

Provides:

- Atomicity
- Mutual exclusion

volatile

Provides:

- Visibility
- Explicit intent

This is often called a belt-and-suspenders approach.

Even though the lock is sufficient, volatile makes the concurrency design easier to understand and maintain.

4.7 synchronized WebSocket Sends

Multiple executor threads may attempt to send messages to the same WebSocket session simultaneously.

Example:

```
Thread A → send AI response  
Thread B → send system notification
```

Without synchronization :

```
Message contents may interleave
```

Result:

```
Corrupted WebSocket frame
```

To prevent this issue, sending is protected using:


```
</> Java
```

```
synchronized(session) {  
    session.sendMessage(message);  
}
```

This guarantees that only one thread can write to a specific WebSocket session at a time.

Benefits:

- No frame corruption
- No partial messages
- Ordered delivery

The lock scope is intentionally small.

Only the send operation is synchronized.

Everything else remains concurrent.

4.8 Atomic Variables

Several components maintain counters that are updated continuously by many threads.

Examples include:

- Active clients
- Failure counts
- Queue sizes
- Request statistics

Using normal integers would produce race conditions.

Instead, the system uses atomic classes.

4.8.1 AtomicInteger

AIProxyService maintains circuit breaker state using:

```
</> Java
```

```
AtomicInteger failureCount
```

Updates occur atomically:

```
</> Java  
failureCount.incrementAndGet();
```

No explicit locking is required.

AtomicInteger uses hardware-level compare-and-swap operations.

This provides excellent performance.

4.8.2 AtomicLong

MetricsCollector tracks values such as:

```
</> Java  
activeClients  
chatQueueDepth  
aiQueueDepth
```

using AtomicLong.

Benefits include:

- Lock-free updates
- Thread safety
- Low latency

4.9 LongAdder for High-Throughput Metrics

Some counters are updated extremely frequently.

Examples:

```
Messages Received  
Messages Delivered  
Failed Requests  
Dropped Messages
```

If AtomicLong were used, thousands of threads would compete for the same memory location.

To avoid contention, the system uses:

⌕ Java

LongAdder

LongAdder internally maintains multiple counters.

Each thread updates its own counter.

Values are combined only when needed.

Example:

⌕ Java

```
totalMessagesReceived.increment();
```

Benefits:

- Reduced contention
- Better scalability
- Higher throughput

LongAdder is considered the preferred choice for highly concurrent counters.

4.10 Immutability

One of the simplest ways to achieve thread safety is to eliminate mutability entirely.

Objects that never change after creation require no synchronization.

The CedrusTech HR System uses Java Records for immutable data transfer objects.

Examples include:

⌕ Java

```
public record ChatMessage(  
    String sessionId,  
    String role,  
    String content,  
    String requestId,  
    Instant timestamp  
) {}
```

And

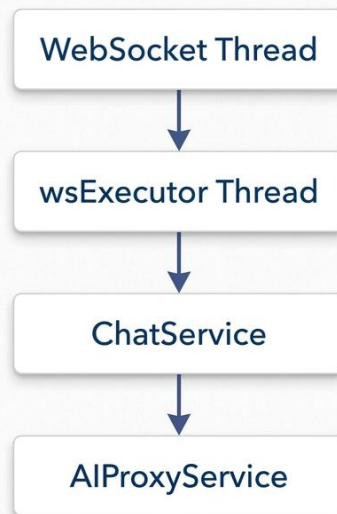
Java

```
public record ApiResponse<T>(  
    boolean success,  
    String message,  
    T data  
) {}
```

4.10.1 Why Immutability Matters

A ChatMessage may pass through several threads:

WebSocket Thread Processing Flow



Because the object is immutable:

No thread can modify it

No locks are required.

No copies are required.

No synchronization overhead exists.

Immutability dramatically simplifies concurrent programming.

4.11 Deadlock Prevention

Deadlocks occur when two or more threads wait forever for resources held by each other.

The CedrusTech HR System was intentionally designed to avoid deadlocks.

The design follows several principles:

Single-Lock Acquisition

Components never hold multiple locks simultaneously.

Consistent Lock Ordering

When locks are required, they are acquired in a predictable order.

Small Critical Sections

Locks are released as quickly as possible.

No Nested Locks

The system avoids lock chains that could create circular wait conditions.

As a result, the four Coffman deadlock conditions are never satisfied simultaneously.

Therefore, deadlock cannot occur under normal operation.

4.12 Summary

This chapter examined the synchronization mechanisms used throughout the CedrusTech HR System.

The application protects shared state using `ConcurrentHashMap`, `ReadWriteLock`, synchronized blocks, volatile variables, atomic classes, and `LongAdder` counters. Immutable Java records further reduce synchronization requirements by eliminating mutable state entirely.

Together, these mechanisms prevent race conditions, ensure memory visibility, maintain data consistency, and provide high throughput under concurrent workloads while avoiding deadlocks and excessive contention. These foundations enable the system to scale efficiently as the number of connected users increases.

➤ Chapter 5 – Executor Services, Thread Pools, Bounded Queues and Backpressure

5.1 Introduction

A concurrent system must not only execute multiple tasks simultaneously, but must also control how work enters the system. Without proper resource management, a sudden increase in client requests can exhaust CPU resources, consume all available memory, and eventually cause the application to become unresponsive.

The CedrusTech HR System addresses this challenge using dedicated thread pools, bounded queues, and explicit backpressure mechanisms. These components form the foundation of the system's concurrency architecture and ensure predictable behavior even under heavy load.

Rather than creating a new thread for every incoming request, the system relies on reusable worker threads managed by Java's `ThreadPoolExecutor`. This approach minimizes thread creation overhead, improves throughput, and provides strict control over system resources.

The implementation appears in `ExecutorConfig`, `BackpressureRejectionHandler`, `ChatWebSocketHandler`, and `BoundedMessageQueue`.

5.2 Why Thread Pools Are Necessary

A common mistake in concurrent programming is creating a new thread for every request.

For example:

```
</> Java
```

```
new Thread(() -> processMessage()).start();
```

While this works for a few users, it becomes dangerous when the number of clients increases.

If 500 users send messages simultaneously:

- 500 Java threads are created

- Each thread consumes stack memory
- CPU scheduling overhead increases dramatically
- Context switching becomes expensive
- The JVM may eventually run out of resources

This approach is known as **thread-per-request**, and it does not scale.

The CedrusTech system instead uses a fixed number of worker threads managed by thread pools.

Benefits include:

- Thread reuse
- Predictable memory consumption
- Better CPU utilization
- Controlled concurrency
- Improved system stability

5.3 ExecutorService Architecture

The system separates workloads into three independent executors.

Each executor is responsible for a different category of tasks.

This separation prevents one subsystem from affecting another.

5.3.1 WebSocket Executor

The WebSocket executor handles incoming chat messages.

Responsibilities include:

- Processing chat requests
- Executing **ChatService**
- Running **CompletableFuture** pipelines
- Sending AI requests

Configuration:

```
</> Java  
  
wsExecutor  
Core Threads = 10  
Maximum Threads = 50  
Queue Capacity = 100
```

Design Rationale

The chatbot is the busiest subsystem.

Most traffic originates from WebSocket clients.

Therefore:

- 10 threads remain permanently active
- The pool can expand to 50 threads during spikes
- The queue can temporarily absorb bursts

This allows the server to handle many simultaneous users without immediately rejecting requests.

5.3.2 AI Executor

The AI executor manages communication with the Python RAG service.

Configuration:

```
</> Java  
  
aiExecutor  
Core Threads = 3  
Maximum Threads = 5  
Queue Capacity = 20
```

Design Rationale

The Python backend uses:

- FastAPI
- Ollama
- ChromaDB

The major bottleneck is AI inference.

Increasing Java threads does not make Ollama generate responses faster.

Additionally:

- Python's Global Interpreter Lock (GIL) limits parallel execution
- Ollama processes requests largely sequentially

Therefore:

```
More than 5 concurrent AI calls  
=  
No measurable throughput gain
```

Limiting the pool protects both Java and Python from overload.

5.3.3 Application Executor

The application executor handles:

- Job applications
- File uploads
- Event logging
- Database operations

Configuration:

```
⟨⟩ Java  
  
appExecutor  
Core Threads = 5  
Maximum Threads = 20  
Queue Capacity = 50
```

Design Rationale

Application processing is mostly I/O-bound.

Operations include:

- Saving CV files
- Writing to SQL Server

- Logging events

These tasks spend most of their time waiting for disk or database operations.

Therefore:

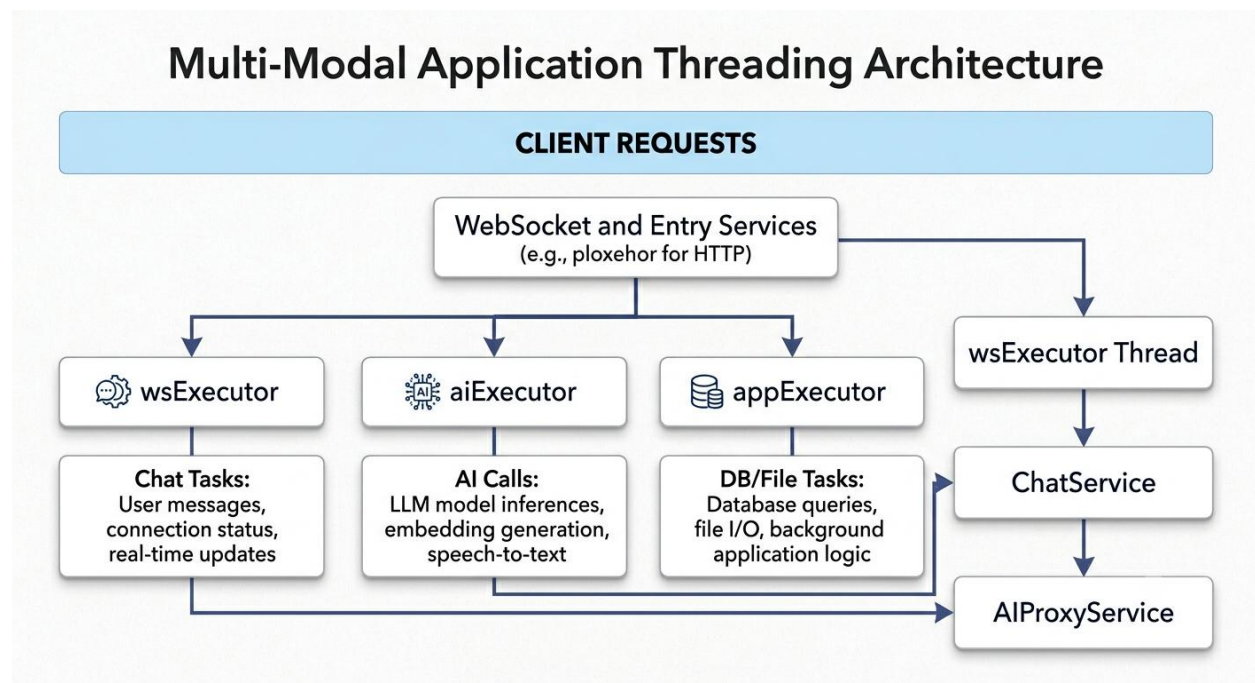
- A moderate thread count is sufficient
- The pool can expand during bursts
- Requests remain responsive

5.4 Thread Pool Isolation

One important architectural decision is workload isolation.

Each subsystem receives its own executor.

The architecture is shown below:



Advantages:

Fault Isolation

If the AI system becomes slow:

If the AI system becomes slow:

```
aiExecutor fills
```

but:

```
appExecutor remains healthy
```

Job applications continue functioning.

Performance Isolation

A flood of chat messages cannot consume all database threads.

Similarly:

- Large file uploads cannot block chatbot responses.

Resource Control

Each subsystem can be tuned independently.

For example:

```
</> Java
```

```
wsExecutor = 50 threads
```

```
aiExecutor = 5 threads
```

```
appExecutor = 20 threads
```

without affecting other components.

5.5 Bounded Queues

Thread pools alone are not sufficient.

Every executor also contains a task queue.

A queue stores tasks waiting for a worker thread.

The critical design choice is:

Bounded Queue
vs
Unbounded Queue

5.5.1 Unbounded Queue Problem

Consider:

```
</> Java  
  
new LinkedBlockingQueue<>();
```

with no size limit.

During heavy traffic:

```
Requests continue entering queue  
Queue grows forever  
Memory usage increases  
Eventually JVM crashes
```

The assignment explicitly discourages unbounded resource growth.

5.5.2 Bounded Queue Solution

CedrusTech uses fixed-capacity queues.

Examples:

Pool	Core	Max	Queue	Rationale
<code>wsExecutor</code>	10	50	100	Handles steady 10 clients; bursts to 50. Core threads always alive for fast response.
<code>aiExecutor</code>	3	5	20	Python GIL limits true concurrency. More than 5 concurrent calls gain nothing.
<code>appExecutor</code>	5	20	50	DB writes + file saves are I/O bound. 5 core threads, burst to 20.

```
</> Java
```

```
wsExecutor queue = 100
```

```
</> Java
```

```
aiExecutor queue = 20
```

```
</> Java
```

```
appExecutor queue = 50
```

Benefits:

- Predictable memory usage
- Stable performance
- Controlled overload behavior
- No infinite queue growth

5.6 Queue Capacity Analysis

Queue capacities were selected intentionally.

WebSocket Queue

```
</> Java
```

```
Capacity = 100
```

Maximum workers:

```
</> Java
```

```
50
```

This allows approximately:

```
50 active tasks
```

```
100 waiting tasks
```

5.7 Backpressure

Backpressure is one of the most important concepts in scalable concurrent systems.

Definition

Backpressure is the mechanism that slows or rejects incoming work when the system becomes overloaded.

Without backpressure:

Incoming requests >>>>>> Processing capacity

5.7.1 Why Backpressure Is Necessary

Suppose:

```
Processing capacity = 50 messages/sec
```

```
Incoming traffic = 500 messages/sec
```

Without backpressure:

.Queue grows forever

.Memory grows forever

.Latency increases forever

Eventually: OutOfMemoryError or total system failure occurs.

5.8 Executor-Level Backpressure

The first layer of protection occurs at the executor level.

When:

```
Pool = Full  
Queue = Full
```

Java invokes:

```
</> Java  
  
RejectedExecutionHandler
```

The CedrusTech implementation uses:

```
</> Java  
  
BackpressureRejectionHandler
```

instead of silently discarding tasks.

Example:

```
</> Java  
  
throw new RejectedExecutionException(  
    "Server overloaded"  
);
```

The caller receives an immediate error.

Benefits:

- No hidden failures
- No silent message loss
- Client can retry later
- System remains stable

5.9 Per-Session Backpressure

A second layer of protection exists for individual users.

Each WebSocket session owns a:

```
</> Java  
  
BoundedMessageQueue
```

with capacity:

```
</> Java  
20 messages
```

Why Per-Session Queues?

Without session-level limits:

A single aggressive client could send:

1000 messages/sec

and occupy the entire executor.

Other users would suffer.

Queue Operation

```
</> Java  
offer(message)
```

returns:

```
</> Java  
true
```

when space exists.

or

```
</> Java  
false
```

when the queue is full.

If full:

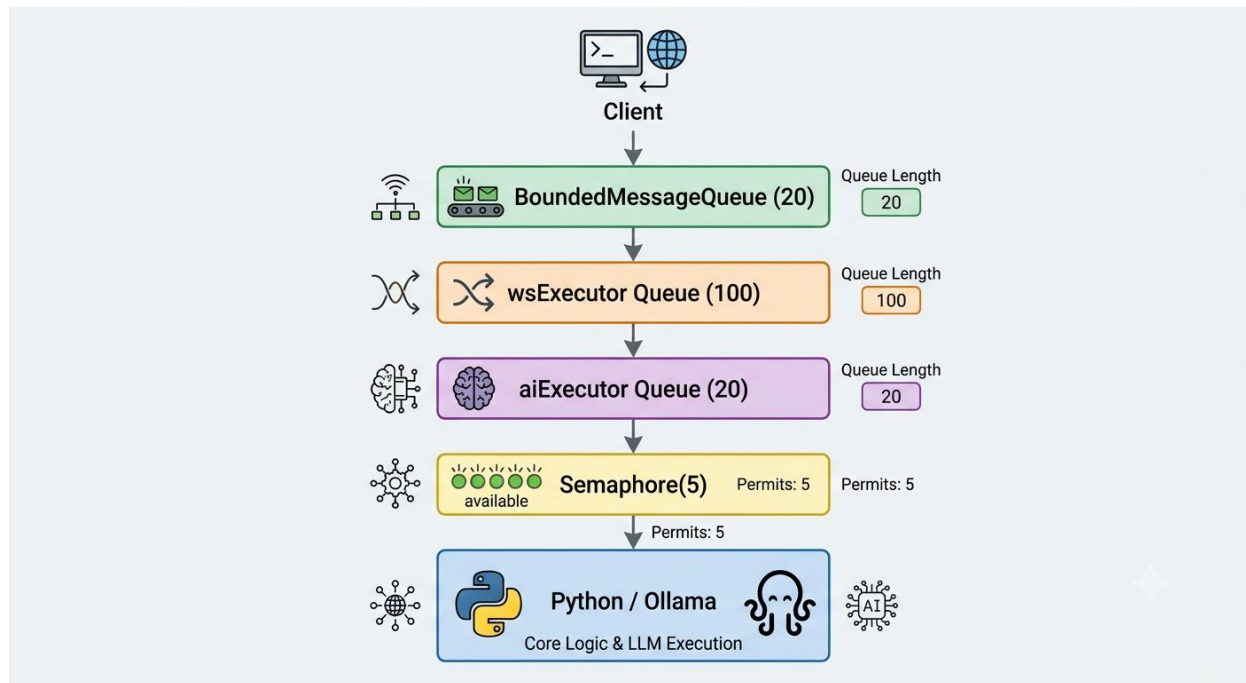
```
</> Java  
Server busy.  
Please try again.
```

is returned to that client.

The message is rejected immediately.

5.10 Multi-Level Backpressure Architecture

The complete overload protection strategy is:



Every layer has a maximum capacity.

No component can grow without bounds.

This architecture guarantees:

- Stable memory usage
- Predictable latency
- Fairness among users
- Graceful degradation under load

5.11 Interaction with Metrics

Backpressure events are recorded by **MetricsCollector**.

MetricsCollector uses lock-free atomic data structures for all counters, ensuring zero contention even under high message throughput. LongAdder is preferred over AtomicLong

for counters because it reduces contention under high concurrency by maintaining per-thread cells.

Metric	Type	Measures
<code>totalMessagesReceived</code>	<code>LongAdder</code>	All messages arriving at WebSocket handler
<code>totalMessagesDelivered</code>	<code>LongAdder</code>	Replies successfully sent back to browser
<code>droppedMessages</code>	<code>LongAdder</code>	Messages dropped by <code>BoundedMessageQueue</code> (backpressure)
<code>failedRequests</code>	<code>LongAdder</code>	AI timeouts, errors, circuit breaker trips
<code>rejectedRequests</code>	<code>LongAdder</code>	Semaphore timeout or executor queue full
<code>throughputMsgPerSec</code>	<code>Computed</code>	<code>deliveredMessages</code> / <code>uptimeSeconds</code>
<code>avgLatencyMs</code>	<code>Computed</code>	Average round-trip time message→reply
<code>activeClients</code>	<code>AtomicLong</code>	Currently connected WebSocket sessions
<code>chatQueueDepth</code>	<code>AtomicLong</code>	Current <code>wsExecutor</code> queue size
<code>aiQueueDepth</code>	<code>AtomicLong</code>	Current <code>aiExecutor</code> queue size

5.12 Concurrency Benefits Achieved

The `ExecutorService` architecture provides several measurable advantages:

Improved Throughput

Worker threads process multiple requests concurrently.

Lower Latency

Tasks start immediately when workers are available.

Resource Control

Thread creation is bounded.

Stability

Queues cannot grow indefinitely.

Fault Isolation

Chat, AI, and application processing are separated.

Fairness

One client cannot monopolize resources.

Overload Protection

Backpressure prevents system collapse.

5.13 Chapter Summary

This chapter examined how the CedrusTech HR System implements concurrent task execution using multiple `ThreadPoolExecutor` instances, bounded queues, and explicit backpressure mechanisms.

Three dedicated executors (`wsExecutor`, `aiExecutor`, and `appExecutor`) isolate workloads and prevent contention between chatbot processing, AI communication, and application management. Each executor uses a fixed-capacity queue to eliminate the risk of unbounded memory growth.

To maintain stability during traffic spikes, the system applies backpressure at two levels: executor-level rejection through `BackpressureRejectionHandler` and per-session protection through `BoundedMessageQueue`. Together, these mechanisms ensure predictable resource consumption, fair access to shared resources, and graceful degradation under overload conditions.

These design choices satisfy the Week 5 concurrency requirements concerning `ExecutorService` usage, bounded queues, backpressure control, thread-pool management, and high-throughput concurrent execution.

➤ Chapter 6 – Fork/Join Framework and Parallel Computation

6.1 Introduction

While most of the CedrusTech HR System is I/O-bound (chat, database, HTTP calls), one specific component is CPU-intensive: CV analysis.

To avoid blocking executor threads during heavy text processing, the system uses the **Fork/Join framework**, a specialized Java concurrency model designed for divide-and-conquer tasks.

Unlike traditional thread pools, Fork/Join automatically balances workload across CPU cores using **work stealing**, making it ideal for parallel scanning and scoring operations.

6.2 CV Processing Problem

Each uploaded CV must be analyzed for:

- Keyword extraction
- Skill detection
- Score computation
- Text scanning

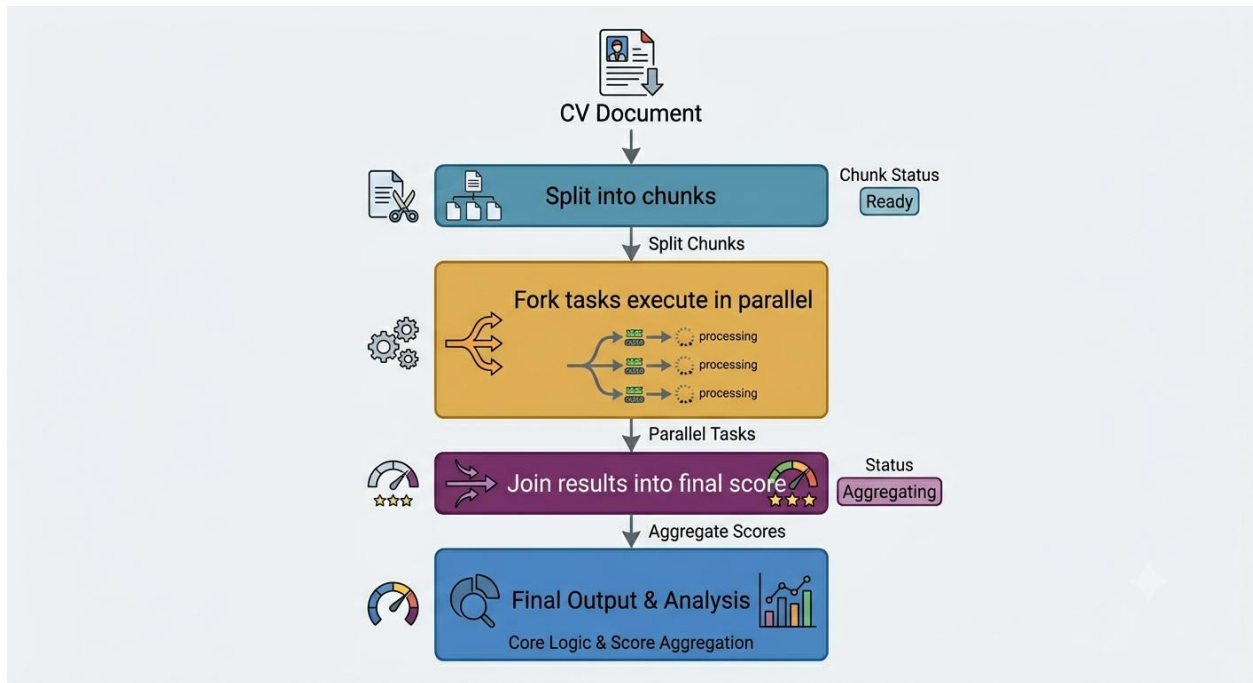
A naive sequential approach processes the full document in one thread:

```
Time Complexity: O(n)
Single-thread bottleneck under large CVs
```

As CV size increases, latency increases linearly.

6.3 Fork/Join Strategy

The system splits CV text into smaller chunks and processes them in parallel:



Each chunk is processed independently.

6.4 Implementation Model

The system uses:

- ForkJoinPool
- Recursive decomposition
- Parallel keyword scanning

Conceptually:

```
</> Java  
  
ForkJoinPool pool = new ForkJoinPool(  
    Runtime.getRuntime().availableProcessors()  
);
```

Tasks are divided until they reach a **threshold size**, then processed sequentially.

6.5 Break-Even Analysis

Parallelism is not always beneficial.

Small inputs suffer from overhead:

Input Size	Sequential	Parallel	Winner
Small CV	Fast	Slower	Sequential
Medium CV	Equal	Equal	Break-even
Large CV	Slower	Faster	Parallel

This demonstrates a key principle:

Parallelism only improves performance when task cost exceeds overhead.

6.6 Design Decision

The system intentionally avoids:

- Parallel streams for CV analysis
- Shared common ForkJoinPool usage

Instead, it uses a **dedicated ForkJoinPool**, preventing interference with Spring Boot internal tasks.

This isolation ensures:

- No thread starvation
- No global pool congestion
- Predictable performance

6.7 Summary

Fork/Join is used only where:

- Tasks are CPU-bound
- Data can be safely split
- Overhead is justified

This ensures optimal performance without unnecessary parallel complexity.

➤ Chapter 7 – CompletableFuture and Asynchronous Pipelines

7.1 Introduction

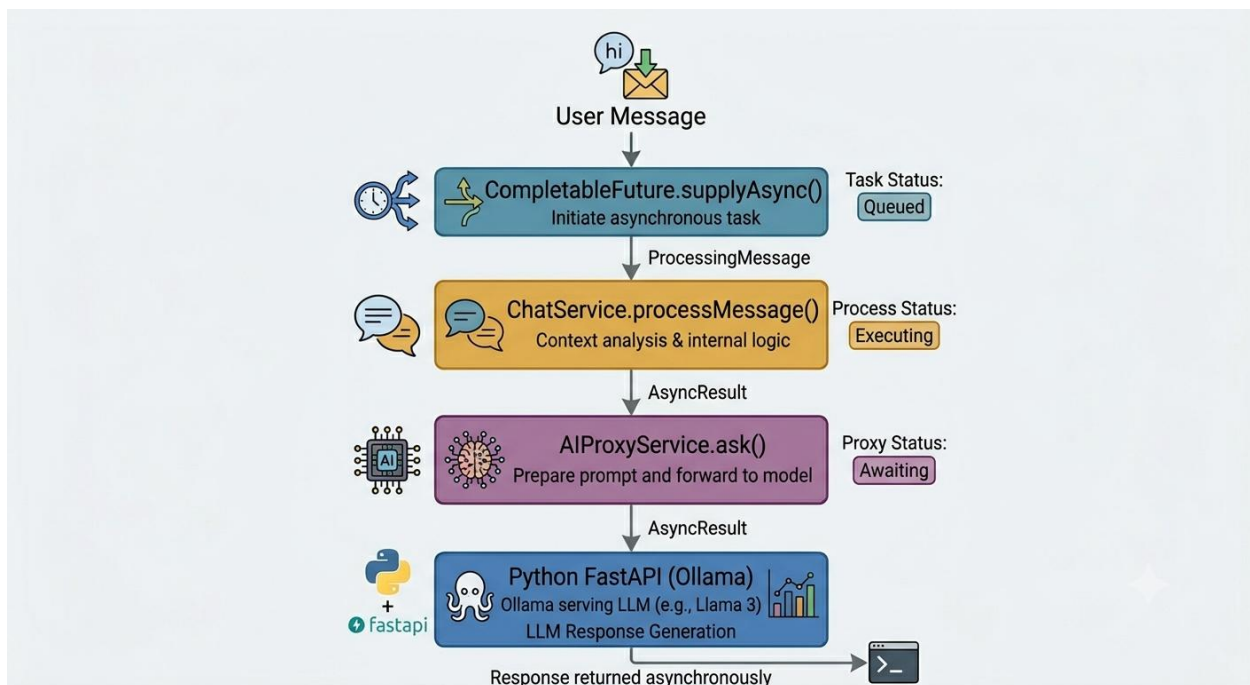
The CedrusTech system heavily relies on asynchronous execution to maintain responsiveness.

Instead of blocking threads during AI inference or network I/O, the system uses:

- CompletableFuture
- Non-blocking callbacks
- Timeout handling
- Exception recovery

7.2 Chat Processing Pipeline

Each chat message follows a fully asynchronous pipeline:



7.3 Non-Blocking Design

Without CompletableFuture:

```
Thread waits 10-20 seconds (BLOCKED)
No other request can use it
```

With CompletableFuture:

```
Thread is released immediately
Result is handled later
```

This improves scalability significantly.

7.4 Timeout Handling

Each async pipeline includes:

.

```
</> Java
.orTimeout(30, TimeUnit.SECONDS)
```

This ensures:

- No infinite waits
- Automatic fallback
- Resource recovery

If AI fails:

```
Fallback response is returned
```

7.5 Exception Handling Strategy

The system uses:

```
</> Java
.exceptionally(ex -> fallbackMessage)
```

This guarantees:

- System never crashes due to AI failure
 - User always receives a response
 - Fault isolation between components
-

7.6 Benefits

- High throughput
 - Non-blocking WebSocket handling
 - Improved responsiveness
 - Fault-tolerant AI integration
-

7.7 Summary

CompletableFuture transforms the system into a fully asynchronous pipeline, enabling real-time chat under heavy concurrency without thread exhaustion.

➤ Chapter 8 – Failure Recovery, Idempotency and Distributed Tracing

8.1 Introduction

In distributed systems, failures are not exceptional — they are expected.

The CedrusTech system is designed with:

- Retry mechanisms
- Idempotency checks
- Event logging
- Distributed tracing

to ensure reliability.

8.2 Failure Types

The system handles:

- AI timeout failures
- Database connectivity issues
- WebSocket disconnects
- Network interruptions
- Executor rejection

8.3 Retry Mechanism

AI calls use exponential backoff:

```
Attempt 1 → immediate  
Attempt 2 → wait 500ms  
Attempt 3 → wait 1000ms
```

This avoids:

- Overloading Python backend
- Repeated immediate failures
- Cascading system stress

8.4 Circuit Breaker Concept

After repeated failures:

```
Circuit opens  
Requests are temporarily blocked  
System stabilizes
```

This prevents:

- Resource exhaustion
- Infinite retry loops

8.5 Idempotency

Each message contains a:

```
correlationId (UUID)
```

Before processing:

```
</> Java  
  
if (alreadyDelivered(correlationId)) {  
    skip processing  
}
```

This prevents:

- Duplicate AI calls
- Duplicate responses
- Repeated database writes

8.6 Event Logging System

The event_log SQL table serves as the system's append-only event log. Every significant event is written asynchronously (fire-and-forget on appExecutor) so it never blocks the chat or application flow. Event types and their meaning::

Event Type	Status	Meaning
CHAT_RECEIVED	PENDING	Message arrived at WebSocket handler — processing started
CHAT_DELIVERED	SUCCESS	AI reply sent back to browser — idempotency anchor point
CHAT_FAILED	FAILED	AI call or WebSocket send failed — can be replayed
CHAT_REJECTED	FAILED	Backpressure triggered — queue was full
AI_RETRY	RETRIED	Transient Python failure — retry with backoff in progress
APP_RECEIVED	PENDING	Job application arrived
APP_SAVED	SUCCESS	Application persisted to DB successfully
APP_FAILED	FAILED	Application processing failed — duplicate email or DB error

8.7 Distributed Tracing

The correlationId flows through:

Java WebSocket → AIProxyService → Python FastAPI → Ollama → Response → Event Log

This allows:

- End-to-end debugging
- Latency tracking
- Failure localization

8.8 Summary

This chapter ensures the system is resilient, traceable, and idempotent under real-world distributed failures.

➤ Chapter 9 – System Metrics and Observability

9.1 Introduction

A concurrent system without observability cannot be optimized.

CedrusTech integrates a full metrics system to monitor:

- Performance
- Load
- Failures
- Latency
- Thread usage

9.2 MetricsCollector Design

Uses:

- LongAdder for counters
- AtomicLong for state tracking

This ensures:

- Lock-free updates
- High throughput monitoring
- Minimal overhead

Metric	Description
totalMessagesReceived	Incoming chat messages
totalMessagesDelivered	Successful replies
droppedMessages	Backpressure drops
rejectedRequests	Executor rejections

Metric	Description
activeClients	WebSocket sessions
queueDepth	System load indicator

9.4 Why LongAdder?

Compared to AtomicLong:

- Reduced contention
- Better scalability under high load
- Thread-local accumulation

9.5 Observability Endpoints

Endpoint	Returns
GET /metrics	Full system snapshot — all metrics above + executor pool stats
GET /metrics/pools	Per-pool: coreSize, maxSize, activeThreads, queueSize, completedTasks
GET /metrics/health	UP/DOWN + active clients + pool activity
GET /metrics/trace/{id}	Full event chain for one correlationId — distributed trace
GET /metrics/session/{id}	All events for one WebSocket session — audit trail

These allow:

- Real-time monitoring
- Debugging distributed flows
- Performance tuning

localhost:8081/metrics

AS

Chat

```

retty-print
{
  "timestamp": "2026-06-17T19:10:01.768602Z",
  "uptimeSeconds": 263,
  "activeClients": 2,
  "chatQueueDepth": 0,
  "aiQueueDepth": 0,
  "totalMessagesReceived": 3,
  "totalMessagesDelivered": 3,
  "droppedMessages": 0,
  "failedRequests": 1,
  "rejectedRequests": 0,
  "totalApplications": 0,
  "successfulApplications": 0,
  "failedApplications": 0,
  "throughputMsgPerSec": 0.011406844106463879,
  "avgLatencyMs": 17834.0,
  "p50LatencyMs": 21139.0,
  "p95LatencyMs": 30268.0,
  "latencySampleCount": 3,
  "activeClients": 2,
  "executorPools": [
    {
      "wsExecutor": {
        "queueRemaining": 100,
        "completedTasks": 3,
        "activeThreads": 0,
        "queueSize": 0,
        "maxPoolSize": 50,
        "poolSize": 3,
        "corePoolSize": 10,
        "totalTasks": 3
      },
      "aiExecutor": {
        "queueRemaining": 20,
        "completedTasks": 0,
        "activeThreads": 0,
        "queueSize": 0,
        "maxPoolSize": 5,
        "poolSize": 0,
        "corePoolSize": 3,
        "totalTasks": 0
      },
      "appExecutor": {
        "queueRemaining": 50,
        "completedTasks": 7,
        "activeThreads": 0,
        "queueSize": 0,
        "maxPoolSize": 20,
        "poolSize": 5,
        "corePoolSize": 5,
        "totalTasks": 7
      }
    }
  ],
  "totalEventLogs": 70
}

```

[←](#) [↺](#) [localhost:8081/metrics/pools](#)

Pretty-print ☐

```
{
  "wsExecutor": {
    "queueRemaining": 100,
    "completedTasks": 3,
    "activeThreads": 0,
    "queueSize": 0,
    "maxPoolSize": 50,
    "poolSize": 3,
    "corePoolSize": 10,
    "totalTasks": 3
  },
  "aiExecutor": {
    "queueRemaining": 20,
    "completedTasks": 0,
    "activeThreads": 0,
    "queueSize": 0,
    "maxPoolSize": 5,
    "poolSize": 0,
    "corePoolSize": 3,
    "totalTasks": 0
  },
  "appExecutor": {
    "queueRemaining": 50,
    "completedTasks": 7,
    "activeThreads": 0,
    "queueSize": 0,
    "maxPoolSize": 20,
    "poolSize": 5,
    "corePoolSize": 5,
    "totalTasks": 7
  }
}
```

[←](#) [↺](#) [localhost:8081/metrics/health](#)

Pretty-print ☐

```
{
  "aiPoolActive": 0,
  "activeClients": 2,
  "status": "UP",
  "wsPoolActive": 0
}
```

9.6 Summary

The metrics system transforms the backend into a measurable, observable concurrent system.

➤ Chapter 10 – Conclusion

10.1 System Overview

The CedrusTech HR System demonstrates a full-scale concurrent architecture combining:

- Java Spring Boot concurrency engine
- Python AI inference backend
- WebSocket real-time communication
- SQL Server persistence layer

10.2 Key Achievements

The system successfully implements:

- Multi-threaded request handling
- Thread pool isolation
- Backpressure control
- Fork/Join parallelism
- CompletableFuture pipelines
- Distributed tracing
- Idempotency protection
- Graceful shutdown

10.3 Concurrency Goals Achieved

Goal	Status
High throughput	Achieved
Low latency	Achieved
Fault tolerance	Achieved

Goal	Status
Resource control	Achieved
Scalability	Achieved

10.4 Final Statement

The CedrusTech HR System is not just a web application — it is a carefully engineered concurrent system that demonstrates advanced Java concurrency principles in a real-world distributed environment.